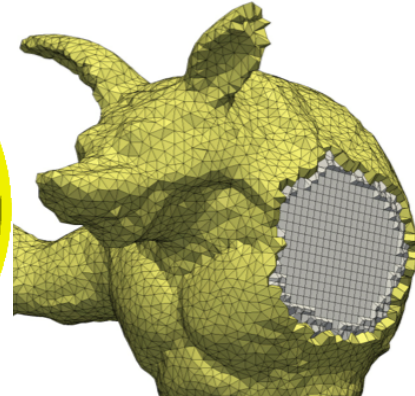
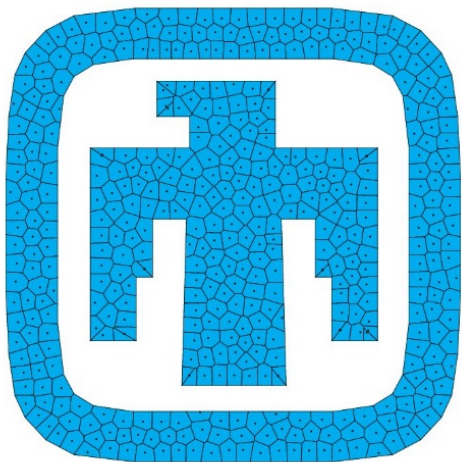


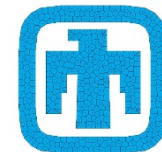
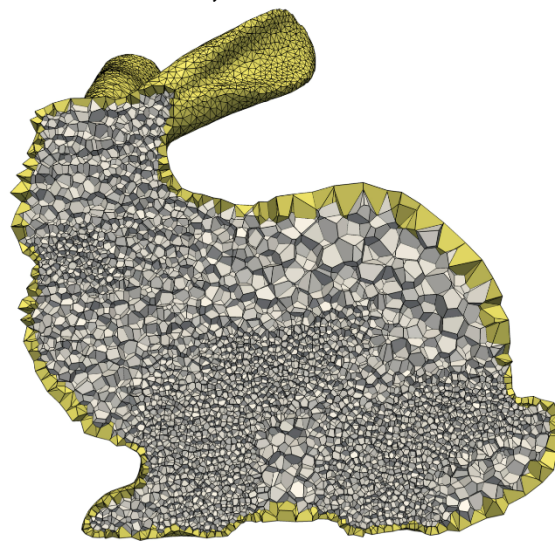
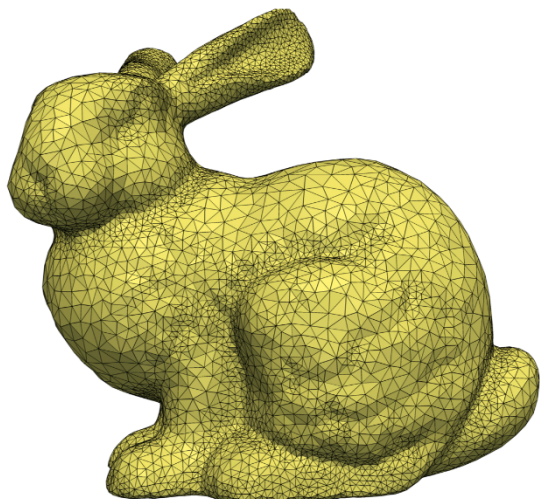


VoroCrust Geometry: 3D polyhedral meshing with true Voronoi cells

conforming to prescribed surface points.

Ahmad Abdelkader Abdelrazek, Univ Maryland,
(speaker instead of Scott)

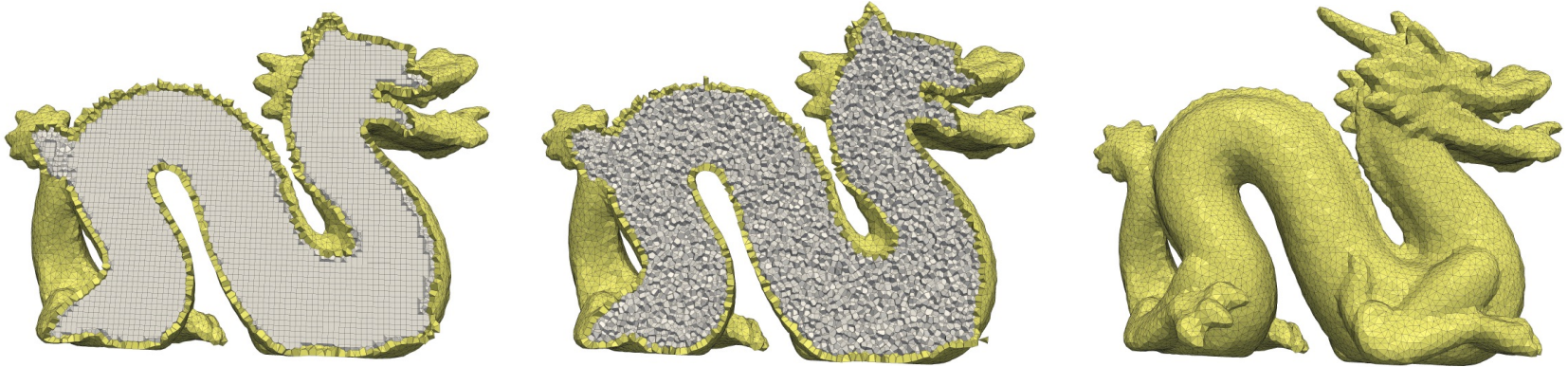
Ahmed H. Mahmoud, Ahmad A. Rushdi, Scott A. Mitchell,
Chandrajit L. Bajaj, John D. Owens, Mohamed S. Ebeida



**Sandia
National
Laboratories**

Related Prior USNCCM Talk

- Mohamed's talk covered algorithm, reconstruction properties, implementation, and example meshes
- This talk covers geometric primal-dual math, algorithmic correctness



VOROCRUST

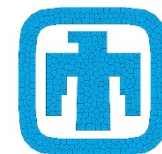
**Volumetric Voronoi Meshing
that Conforms to Surface Samples**

- Mohamed Ebeida
- Optimization and Uncertainty Quantification Department
- Sandia National Laboratories



VoroCrust

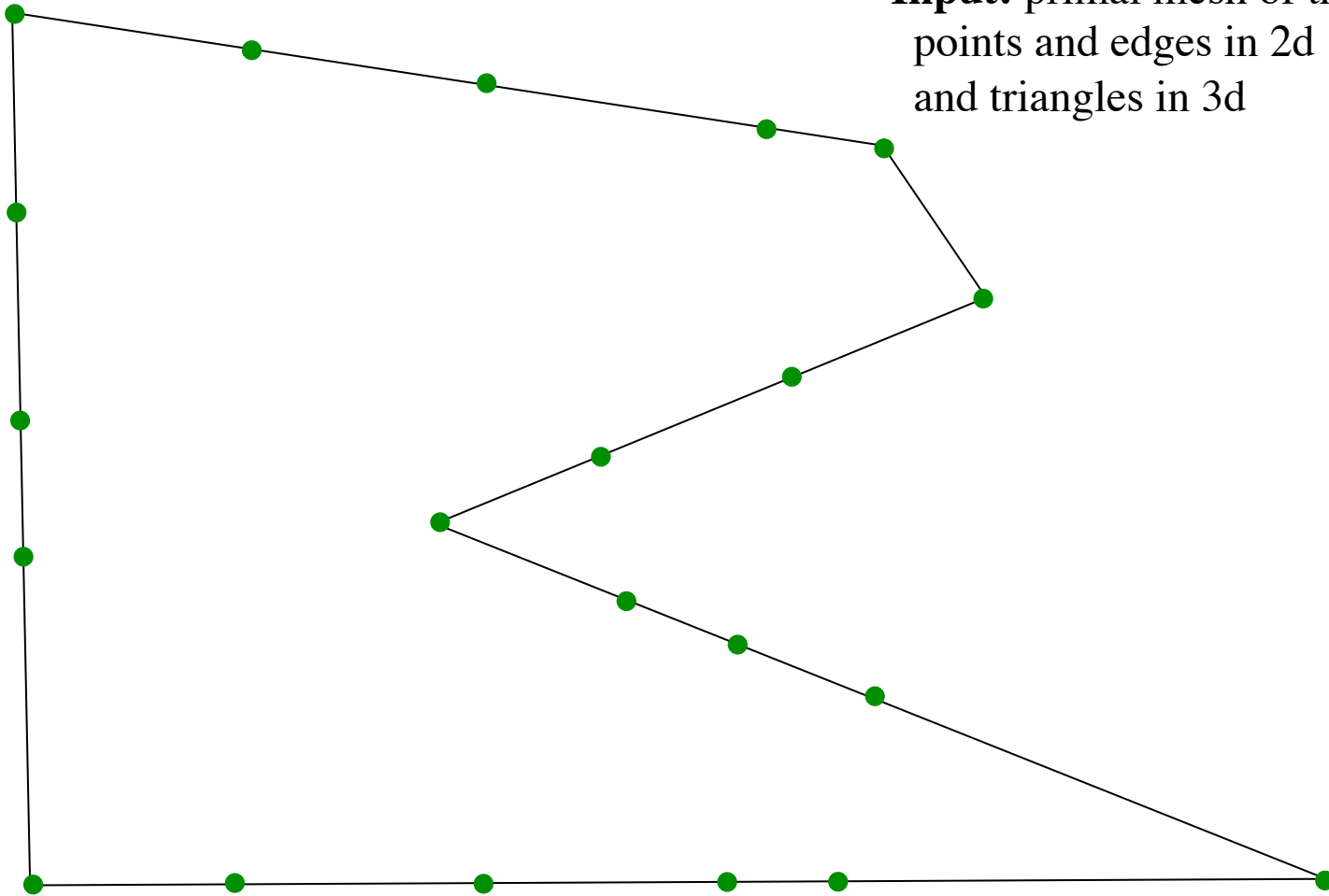
ALGORITHM OVERVIEW



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'Classic' VoroCrust Primal-Dual-Primal Dance

Input: primal mesh of the input surface,
points and edges in 2d
and triangles in 3d

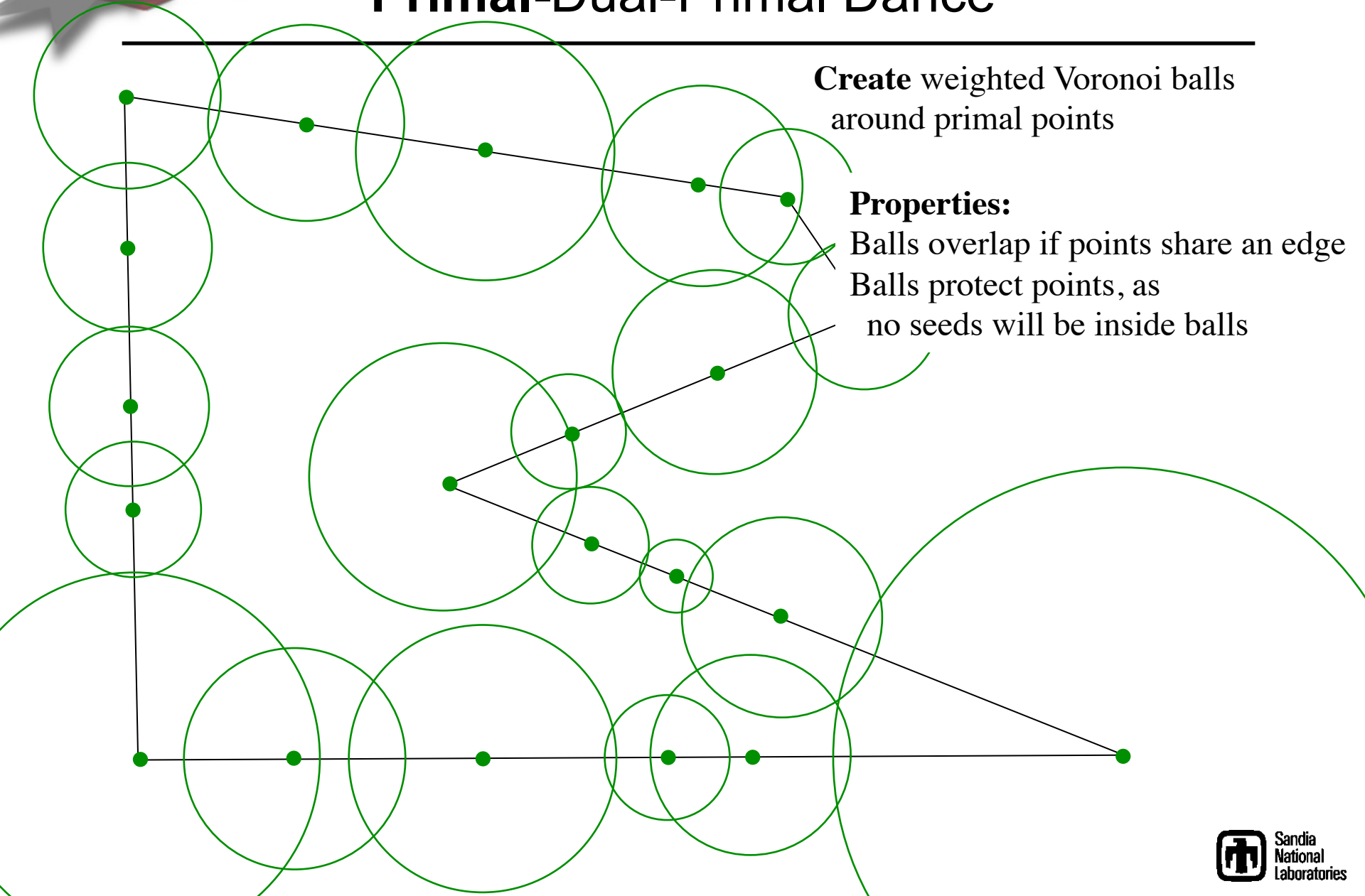


Primal-Dual-Primal Dance

Create weighted Voronoi balls
around primal points

Properties:

Balls overlap if points share an edge
Balls protect points, as
no seeds will be inside balls



Primal-Dual-Primal Dance

Create dual unweighted seed pairs at weighted Voronoi sphere intersections

red = exterior

blue = interior

2D: 2 circles intersect at two points

3D: 3 spheres intersect at two points

Require: both intersection points lie outside all other balls

Want seeds sufficient to reconstruct surface
surface mesh nodes are Voronoi vertices
surface mesh edges are Voronoi edges
surface mesh triangles are Voronoi facets

Primal-Dual-Primal Dance

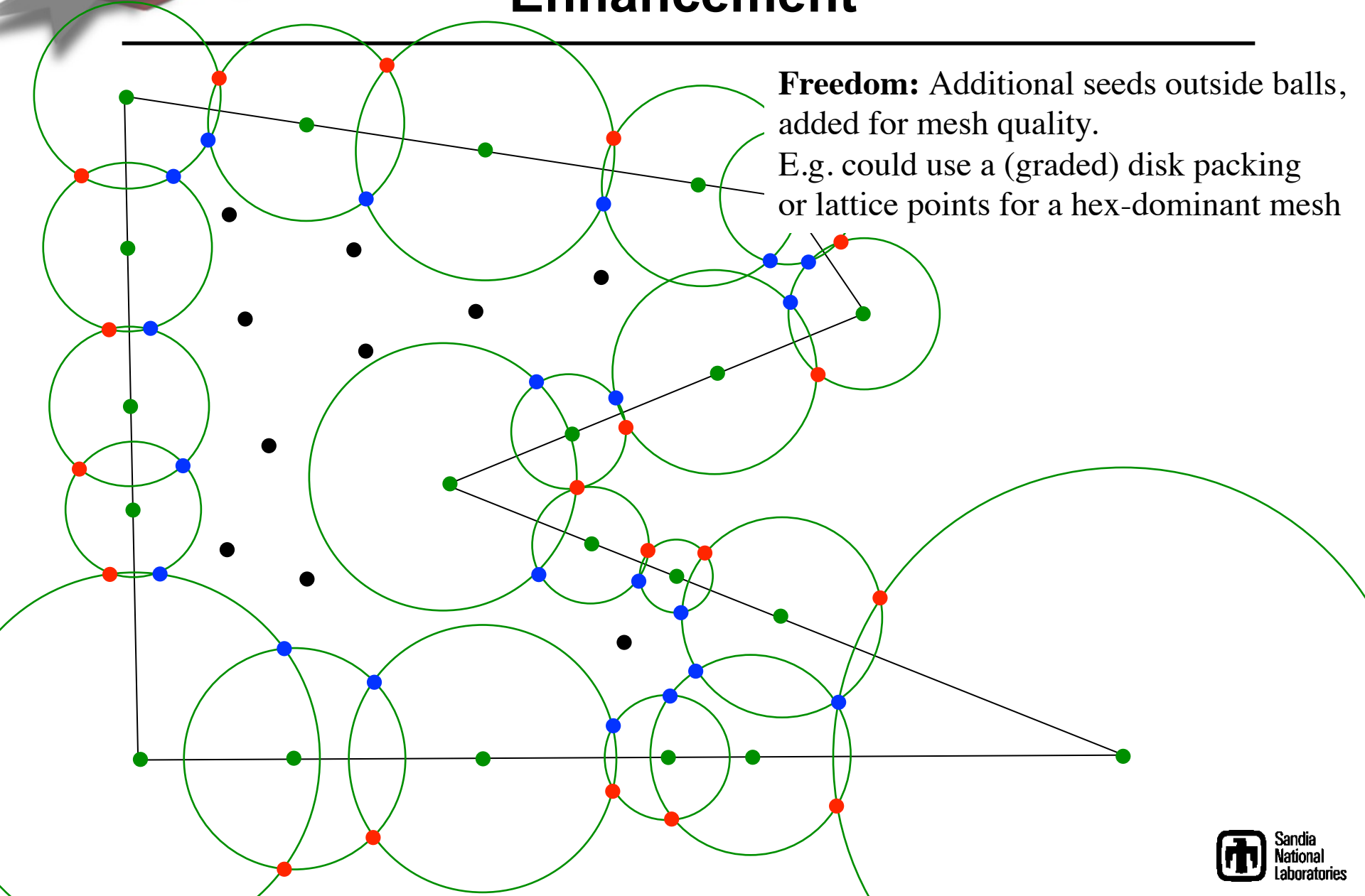
Property: Unweighted Voronoi diagram reconstructs the surface mesh

Want seeds sufficient to reconstruct surface
surface mesh nodes are Voronoi vertices
surface mesh edges are Voronoi edges
surface mesh triangles are Voronoi facets

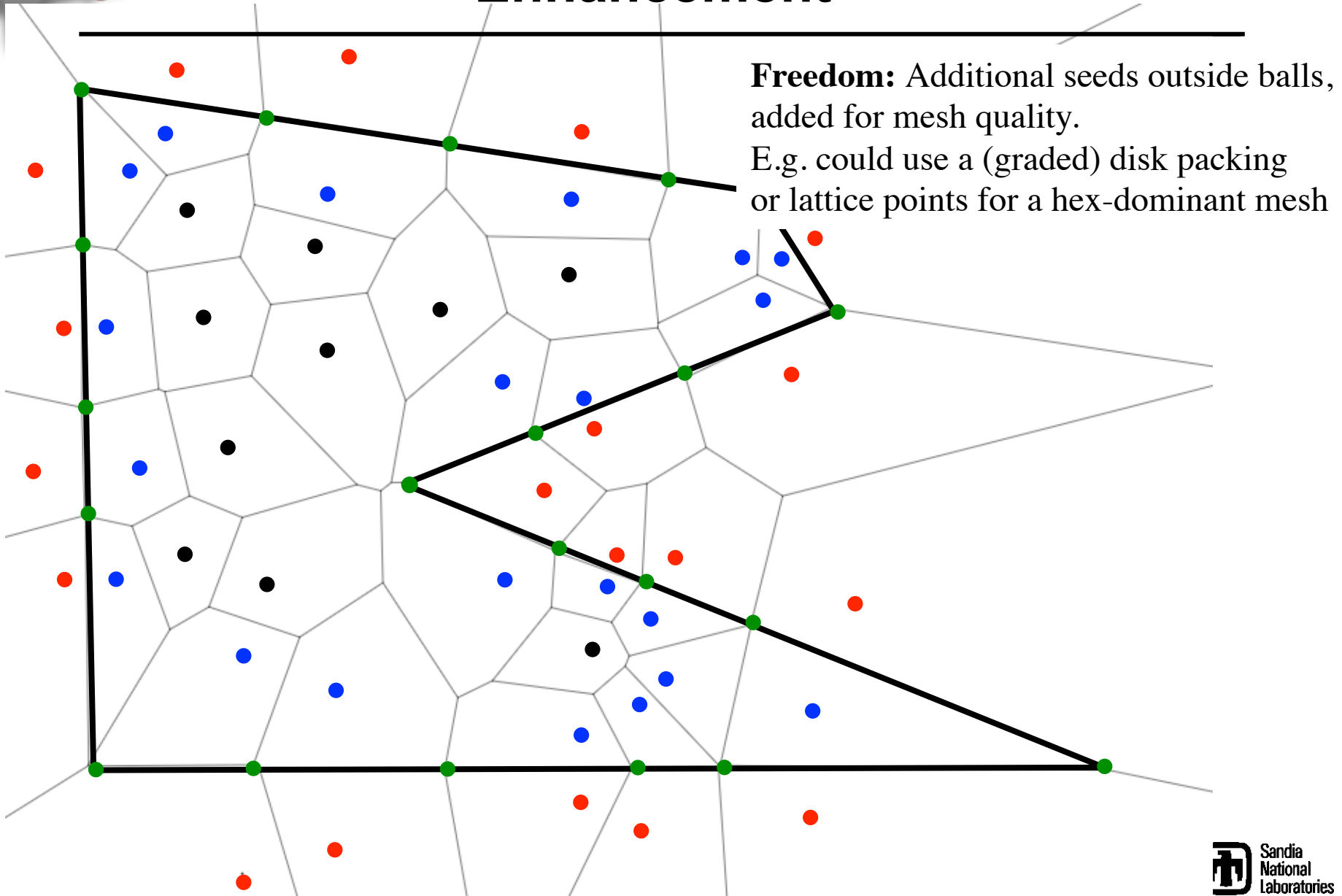
Enhancement

Freedom: Additional seeds outside balls, added for mesh quality.

E.g. could use a (graded) disk packing or lattice points for a hex-dominant mesh



Enhancement





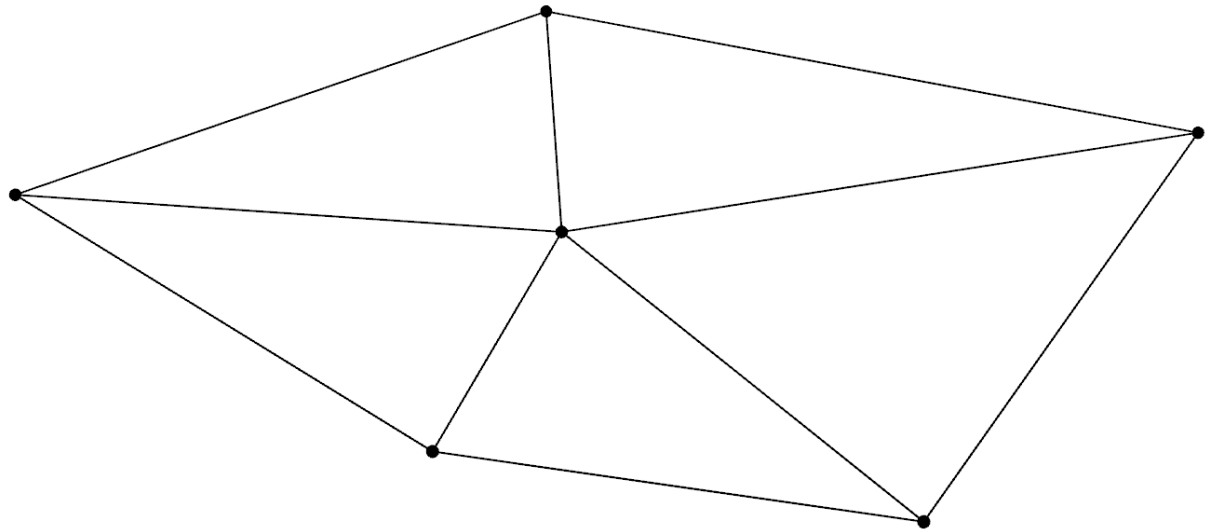
Challenge

- **Problem:**
 - Sometimes one intersection point is covered, and the other in the pair is uncovered
- **Solution:**
 - “Basic” algorithm adds additional seeds
 - reconstructs vertices, edges, facets one by one
 - reconstructs a refinement of the input mesh



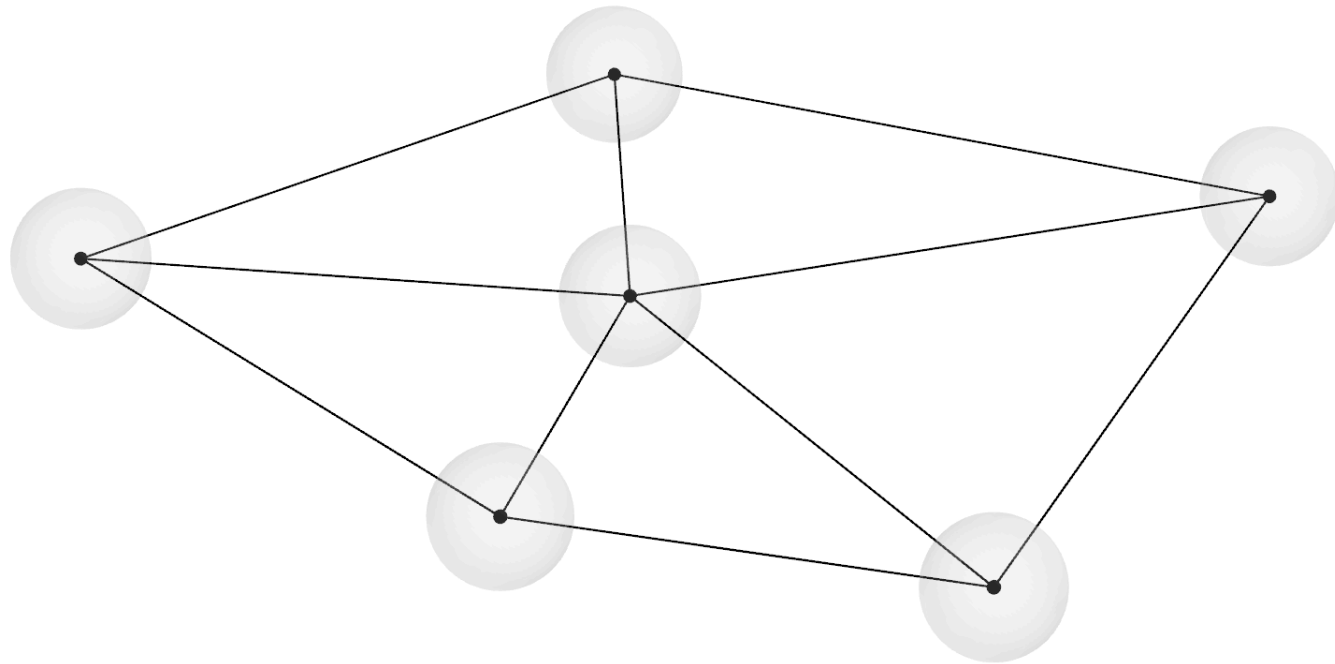
Basic VoroCrust Algorithm

- **Step (0): input mesh**



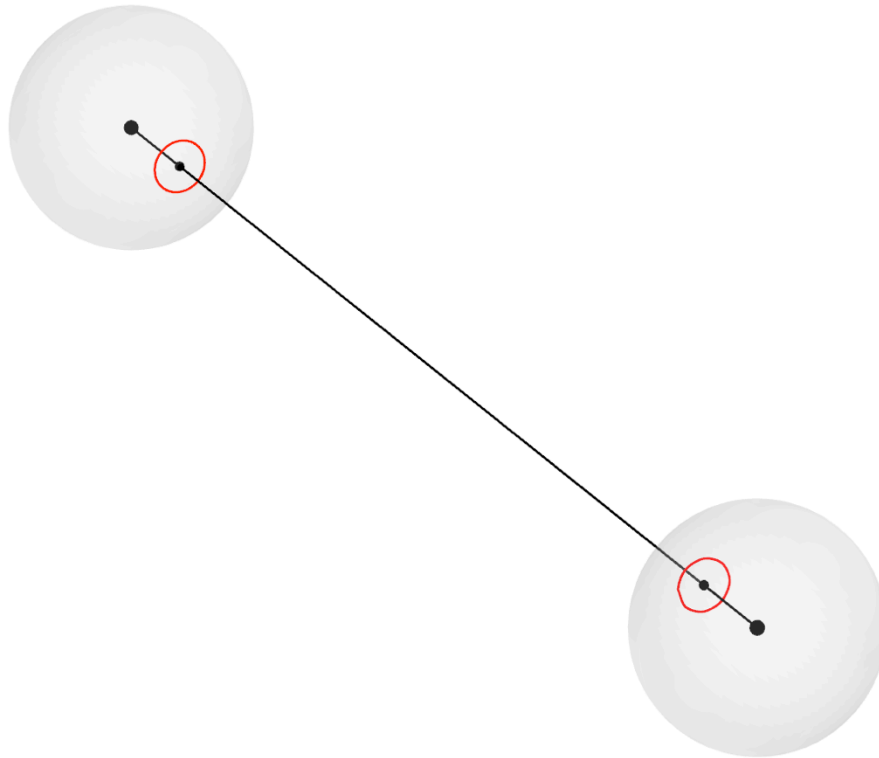
Basic VoroCrust Algorithm

- **Step (1): Spheres around each mesh node**



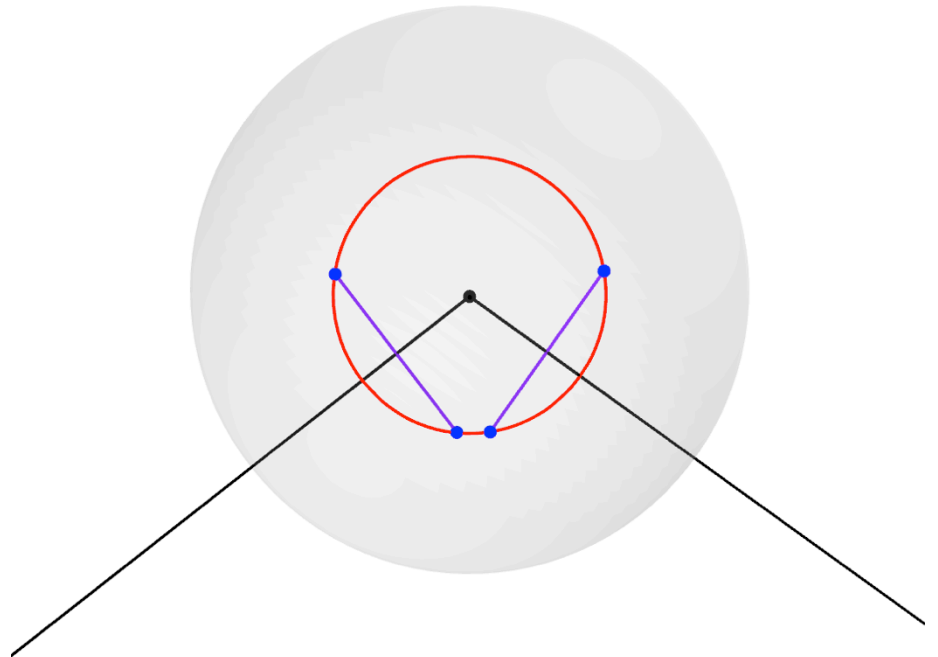
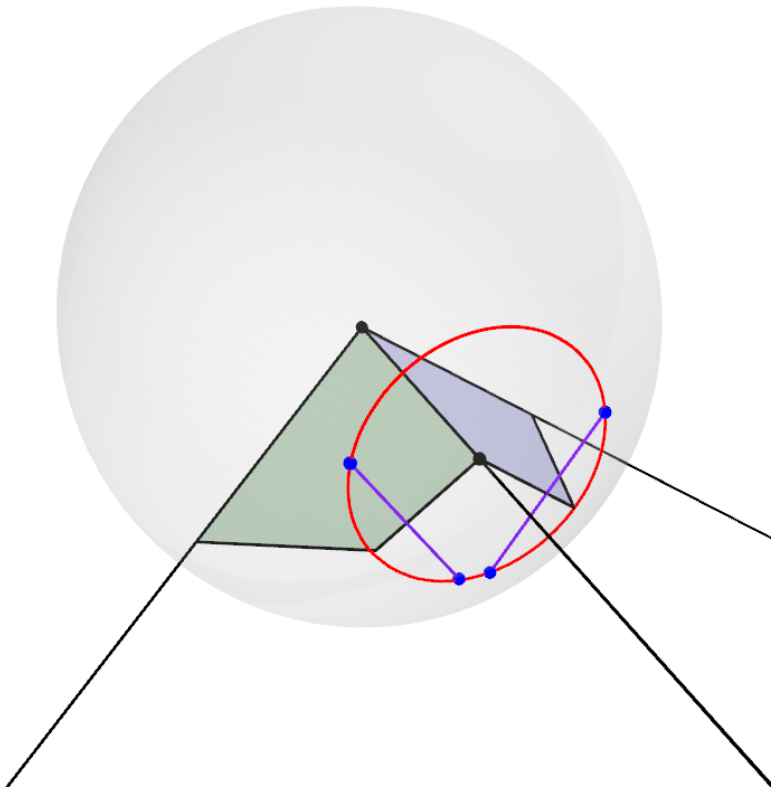
Basic VoroCrust Algorithm

- **Step (2):** Circles around each mesh edge, on sphere



Basic VoroCrust Algorithm

- **Step (3): Four seeds on each circle**





Basic VoroCrust Correctness

- **Reconstruction Properties: Vertices**
- **By Construction ...**



Basic VoroCrust Correctness

- **Reconstruction Properties: Edges**

Claim 1. $\forall p \in \overline{ab}, \forall v \notin \{a, b\}, \min\{d(p, a), d(p, b)\} < d(p, v).$



Basic VoroCrust Correctness

- **Reconstruction Properties: Faces**

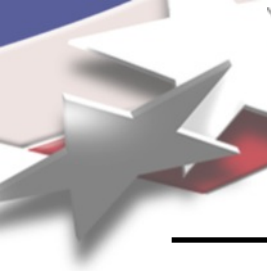
Claim 2. $\forall p \in \triangle abc, \forall v \notin \{a, b, c\}, \min\{d(p, a), d(p, b), d(p, c)\} < d(p, v).$



Conclusion

- **Work in progress**
- **Status**
- **Open**

- 
-
- **backup slides**



red: 58, 218, 75, 329, 53, 432, 74, 509.5, 204, 718, 268, 699
blue 122, 218, 105, 329, 129, 430, 112, 509, 203, 564, 269, 588

tweaked:

red: 58, 218, 75, 329, 53, 432, 74, 509.5, 204, 718, 268, 699,
blue: 122, 218, 105, 329, 129, 430, 112, 509, 203, 564, 269, 588



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